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## Quantitative Modeling in Finance

**Quantitative finance is a branch of finance that applies mathematical and statistical models to understand financial markets and pricing of financial derivatives.** The concepts and models used in quantitative finance are essential for managing risk, pricing options, and predicting future market behavior.

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### 1. Derivative Pricing by Replication in Binomial Model

The **binomial model** is a simple discrete-time model used to price derivatives, such as options, by simulating the price evolution of the underlying asset over time. This method replicates the payoff of the derivative using a portfolio consisting of the underlying asset and a risk-free asset (such as a bond).

#### Key Steps:

- Construct a **binomial tree** to represent possible price movements over time. **(Always remember there could be only 2 possibilities, either the stock price will go up or go down). Anyway, on the day of expiry the premium of an option will become zero.**
- At each node, calculate the option value by replicating the payoff at the terminal nodes and working backward.
- The risk-neutral probabilities are used to account for risk preferences and price the option.

#### Practice Example:

- Consider a stock that can either move up by 10% or down by 10% in one period. The risk-free rate is 5%. Calculate the price of a European call option with a strike price of 100, expiring in one period.

*Since this PDF file is for concept understanding, the practical solution is given in my practical PDF file.*

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### 2. Discrete Probability Space, Filtration, Conditional Expectation

A **discrete probability space** consists of a finite set of outcomes and their associated probabilities. In financial modeling, this is used to define possible future states of the world and the likelihood of each state.

#### Let's understand this in simple:

Let say you're playing a simple game where you roll a die. The die has six faces, so the **possible outcomes** are {1, 2, 3, 4, 5, 6}. These are **discrete outcomes** because they're limited to just these six numbers—nothing in between. Now, each of these outcomes has a certain **probability**. For example, if the die is fair, each number (1, 2, 3, 4, 5, or 6) has an equal chance of showing up, so the probability of each is  $1/6$ .

In finance, we use a similar idea: we define a set of **possible outcomes** for something like the price of a stock in the future. We also assign a **probability** to each outcome, which tells us how likely each scenario is. For example, the stock could go up, stay the same, or go down, and we'll assign probabilities to each of those possibilities.

**Filtration** refers to the collection of information available up to a particular time. In other words, if we're looking at the stock market, filtration tells us the prices of stocks, interest rates, or other data we have at the current time. This information changes over time as we get new updates.

**Conditional Expectation** is the expected value of a random variable, given some known information (or filtration). This helps in understanding the expected future payoff given the information available at the current time.

Let's understand this in simple:

Imagine you're trying to predict the weather for tomorrow. Normally, you'd just guess, but if you know today's weather (like if it's cloudy), your guess would be more accurate. **Conditional Expectation** is like making that smarter guess based on what you already know. It's the **expected outcome** (or average result) of something, but with the condition that you already have some information.

In finance, this means predicting the future value of an asset (like a stock) based on the information you have right now. For example, if you know that a stock price has been rising for a few days, you might expect it to keep rising tomorrow (but with some uncertainty). The conditional expectation is the expected future payoff, considering the data you have at the current moment.

In simple terms:

- **Discrete Probability Space** is just a set of possible future events and their likelihood.
- **Filtration** is all the information we know at any point in time.
- **Conditional Expectation** is our best guess for the future, based on the information we have at the moment.

**Practice Example:**

- You have a probability space where the stock can either go up by 20% or down by 10%. If the stock price today is 50, calculate the expected price if the probability is 50% on each side, in one period given that you know the stock has gone up.
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### 3. Discrete-Time Martingales

A **martingale** is a stochastic process where the expected future value of the process, given all the past information, is equal to the current value. In finance, martingales are used to model fair pricing of financial assets.

A **discrete-time martingale** is a sequence of random variables where each value is conditionally independent of previous values, given the information up to the current time.

For example:

- If you're pricing a stock, a martingale model would imply that you can't predict whether the stock will go up or down tomorrow based on the price today. On average, **today's price is the best estimate for tomorrow's price**, given everything we know.

This idea is used a lot in models of financial markets, where asset prices behave randomly, and the idea is that **no strategy can consistently outperform the market**.

### What is a Discrete-Time Martingale?

Now, let's talk about **discrete-time martingales**. In simple terms:

- **Discrete-time** just means we're looking at **time in steps**. For example, instead of analyzing prices every millisecond, we might look at prices at the end of each day, week, or month.
- A **discrete-time martingale** is a sequence of events (prices or values) where, at each step, the **expected value of the next outcome is equal to the current outcome**, given the information we have up until now.

### Practice Example:

- Consider a random walk model for the price of a stock where the price changes up by 1 unit or down by 1 unit with equal probability. Verify if this is a martingale.

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## 4. Risk-Neutral Pricing in Binomial Model for European and American Derivatives

**Risk-neutral pricing** is a fundamental concept in modern financial theory, where all assets are assumed to grow at the risk-free rate in a risk-neutral world. This simplifies the pricing of derivatives, as the future price of an asset is discounted at the risk-free rate.

For both **European and American options**, the binomial model can be used to compute prices:

- **European Option:** Can only be exercised at expiration. The binomial model uses backward induction to calculate the option price.
- **American Option:** Can be exercised at any time before expiration. The option price is the maximum of exercising the option or holding it, calculated at each node of the binomial tree.

### Practice Example:

- For a European put option with a strike price of 100, maturity of 1 period, and current price of the underlying asset at 95, use the binomial model to calculate the price assuming up and down factors of 1.1 and 0.9, respectively.

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### Summary of Concepts:

- **Binomial Model:** Used for derivative pricing through replication and backward induction.
- **Discrete Probability Space:** Framework for defining possible outcomes.

- **Filtration & Conditional Expectation:** Concepts to model information and expectations in a dynamic environment.
  - **Martingales:** Processes with no predictable drift.
  - **Markov Chains:** Models where the future depends only on the present state.
  - **Risk-Neutral Pricing:** Pricing derivatives by assuming no risk preferences.
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